

Question Paper

Exam Date & Time: 07-Jan-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

THIRD SEMESTER B.TECH. END SEMESTER DEGREE EXAMINATIONS - DECEMBER 2024 / JANUARY 2025
SUBJECT: ECE 2124/ECE_2124 - DIGITAL SYSTEM DESIGN
(MAKEUP)

Marks: 50

Duration: 180 mins.

Answer all the questions.

- 1A) Design a combinational circuit to convert 4-bit Binary to Gray code. (4)
- 1B) Design the specified Boolean function using 8:1 multiplexer, where D is the input and A, B, C are utilized as the selection lines. (3)
 $F(A,B,C,D) = AB' + BD + B'CD'$
- 1C) Implement 4 to 16 decoder using 2 to 4 decoder using optimized number of decoders. (3)
- 2A) Design a 2-bit magnitude comparator using basic logic gates. (4)
- 2B) Design a mod-6 ripple up counter using negative edge triggered T Flipflops with active high clear. (3)
- 2C) Design a synchronous 3-bit Gray code counter using positive edge triggered T flipflops. (3)
- 3A) Design sequence detector to detect the overlapping sequence 0101 using Mealy model with positive edge triggered D Flipflop. (4)
- 3B) Convert D flipflop to JK flipflop. Write characteristic equations of JK flip flop. (3)
- 3C) Write a structural Verilog for the circuit shown in Fig 3C. (3)

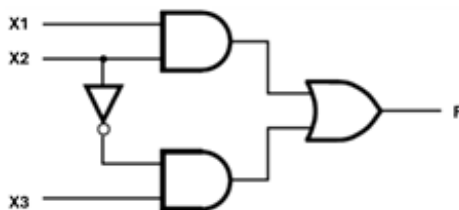


Fig 3C

- 4A) Write a behavioral Verilog code for 2 bit comparator using if else statement. (4)
- 4B) Write a dataflow Verilog code for 2:4 decoder (3)
- 4C) Write a Verilog code for a 3 input XOR gate using user defined primitive (UDP) (3)
- 5A) Implement full adder using ACT 1 logic module (4)
- 5B) Explain ASIC design flow with neat diagram (3)
- 5C) What is TTL logic? Explain the working of 2 input NAND gate using TTL logic (3)

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